

**INTERNSHIP REPORT**

**ON**

**AI DEPLOYMENT & AUTOMATION**

**Submitted To**

**SCHOOL OF COMPUTER SCIENCE AND ENGINEERING**  
**IILM UNIVERSITY, GREATER NOIDA, U.P.**

In partial fulfilment of the requirement of degree of

**B.TECH CSE**

**Submitted By**

**TOSHI PANDEY**

ROLL NUMBER: 2410030422

SECTION: 3CSE3

SEMESTER: 5TH SEMESTER

BATCH: 2026-27

**Internship Organization: EduSkills Foundation**

Program: AICTE - EduSkills Virtual Internship

Internship Duration: June 2026 - August 2026

Academic Session 2026-27

## **1. CANDIDATE'S DECLARATION**

I, TOSHI PANDEY, hereby declare that the internship report titled “AI Deployment & Automation” has been prepared by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering at IILM University, Greater Noida.

This report presents my learning and project-oriented exposure during the 8-week AICTE - EduSkills Virtual Internship Program in the domain of AI Deployment & Automation, undertaken during June 2026 to August 2026. The report has been prepared using the official internship offer letter, completion certificate, and the structured learning plan supplied by EduSkills.

I affirm that the report has been prepared in good faith and that the supporting documents and sources used in its preparation have been acknowledged. I understand that I am responsible for the accuracy of the personal and academic details contained in this report.

Signature: \_\_\_\_\_

Student Name: TOSHI PANDEY

Roll No.: 2410030422

## **2. ACKNOWLEDGEMENT**

I express my sincere gratitude to IILM University, Greater Noida, and the School of Computer Science and Engineering for providing me with the opportunity to undertake a summer internship in the field of AI Deployment & Automation.

I am thankful to EduSkills Foundation and the AICTE - EduSkills Virtual Internship Program for providing a structured eight-week learning experience covering AI deployment foundations, FastAPI model serving, Docker and CI/CD, cloud AI deployment, generative AI serving, AI agents, automation workflows, AI security and compliance, Kubernetes, monitoring and observability, performance optimization, model drift, advanced RAG, and applied AI projects.

I am also grateful to the mentors, instructors, faculty members, and everyone involved in the program for their guidance, assessments, project activities, and learning resources. The internship helped me understand how AI solutions progress from development to deployment and how automation, monitoring, security, and optimization contribute to reliable AI systems.

Finally, I thank my parents, friends, and classmates for their encouragement and support throughout the internship and the preparation of this report.

Signature: \_\_\_\_\_

TOSHI PANDEY

Roll No.: 2410030422

### 3. INTERNSHIP COMPLETION CERTIFICATE

The following page contains the official Internship Credential issued by EduSkills Academy. The certificate confirms successful completion of an 8-week program in the domain of AI Deployment & Automation and records the learner competencies covered during the internship.

**CERTIFICATE  
OF INTERNSHIP**





**PRESENTED TO**

***Toshi Pandey***

*has successfully achieved the Internship Credential by completing a 8-week program in the domain of*

**AI Deployment & Automation**

*During the internship, the learner demonstrated proficiency in:*

- Foundations of AI Deployment & Production Readiness & Advanced API Deployment with FastAPI
- Containerization with Docker & Introduction to MLOps & CI/CD Pipelines
- Cloud Deployment Basics (AWS SageMaker, Google Vertex AI, Azure ML) & Deploying Generative AI Applications
- Building Advanced AI Agents with LangChain & LlamaIndex & AI-Powered Automation Workflows
- Integrating AI into Web/Mobile Apps & Security and Compliance in AI Deployment
- Container Orchestration with Kubernetes & Monitoring, Logging & Observability
- Performance Optimization & Cost Management & Model Drift Detection & Retraining & Advanced RAG & Agentic Workflows
- Case Studies & Best Practices in AI Deployment & Project: Deploy a Full-Stack RAG Application & Project: AI Agent for Business Automation & Capstone Project: End-to-End AI Solution for Business Problem

This certification recognizes the successful completion of the internship requirements.



Issued on: Sep 10, 2026  
Certificate ID: 2026-D1DDA7AB3A



**Mifu Swain**  
General Manager  
(Learning & Development)  
EduSkills

**Certificate details: Issued on September 10, 2026 | Certificate ID: 2026-D1DDA7AB3A**

## **ABSTRACT**

Artificial Intelligence deployment is the engineering process of moving AI models and applications from development environments into usable, repeatable, monitored, and maintainable services. This internship report presents the learning and project-oriented exposure gained during an 8-week AICTE - EduSkills Virtual Internship in the domain of AI Deployment & Automation.

The internship followed a structured week-wise curriculum. The first stages addressed AI lifecycle concepts, production readiness, MLOps terminology, environment setup, and FastAPI model serving. The learning then progressed to containerization with Docker, CI/CD pipelines, cloud AI deployment using Amazon SageMaker, Google Vertex AI, and Azure Machine Learning, and the deployment of Generative AI applications including LLM serving concepts, vLLM, and model quantization.

The later weeks introduced AI agents and automation workflows using LangChain, LlamaIndex, and LangGraph; AI integration using Streamlit and Gradio; security, privacy, compliance, and bias mitigation; Kubernetes-based orchestration and observability with Prometheus and Grafana; performance optimization, cost management, drift detection using Evidently AI, retraining, and advanced RAG architectures with vector databases.

The capstone-oriented learning included a full-stack RAG application, an AI agent for business automation, and an end-to-end AI solution for a business problem. The main outcome of the internship was a practical understanding of how AI deployment, automation, integration, monitoring, and lifecycle management work together as parts of a production-oriented AI system.

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## **4. PROJECT DESCRIPTION**

### **4.1 Introduction**

The internship was completed under the AICTE - EduSkills Virtual Internship Program in the domain of AI Deployment & Automation. The official offer letter identifies the program as an 8-week virtual internship designed around structured learning and practical exposure. It also states that the program included weekly assessments, project documentation or assignments for certain modules, and a Final Assessment Test in the final week.

The main theme of the internship was the engineering lifecycle required to place AI solutions into practical use. Instead of focusing only on model creation, the curriculum addressed model serving, APIs, packaging, CI/CD, cloud deployment, Generative AI serving, agents, application integration, security, orchestration, observability, optimization, model drift, retraining, RAG, and end-to-end project implementation. The official completion certificate confirms successful completion of the internship requirements and records proficiency in the major technical areas taught during the program. The certificate also highlights three project-oriented outcomes: a full-stack RAG application, an AI agent for business automation, and an end-to-end AI solution for a business problem.

### **4.2 Organization Profile**

EduSkills Foundation describes itself as a non-profit organization working to enable an Industry 4.0-ready digital workforce in India. Its stated vision is to bridge the gap between academia and industry by ensuring access to world-class curriculum for faculties and students. EduSkills states that it works with students, faculties, educational institutions, and government stakeholders through skilling interventions and that its impact model is built around Education, Employment, and Entrepreneurship.

The organization presents the AICTE - EduSkills Virtual Internship as a major part of its internship ecosystem. Its internship platform describes a process involving application, verification and offer-letter issuance, an 8-week learning module, project submission and credential validation, final assessment, and internship certification after successful completion.

For this internship, the selected domain was AI Deployment & Automation. The official offer letter identifies Toshi Pandey as a B.Tech student of IILM University, Greater Noida, and records the internship duration as June 2026 - August 2026.

Organization: EduSkills Foundation

Program: AICTE - EduSkills Virtual Internship

Internship Domain: AI Deployment & Automation

Duration: June 2026 - August 2026

Mode: Virtual

Institute: IILM University, Greater Noida

Course: B.Tech (Computer Science and Engineering)

Student: Toshi Pandey

Section: 3CSE3 | Semester: 5th Semester | Roll No.: 2410030422

### **4.3 Problem Statement**

An AI model cannot deliver a complete production solution by itself. A useful AI service must be packaged with its dependencies, exposed through an appropriate interface, integrated with applications, deployed into a suitable environment, observed after release, and maintained as models, data, and dependencies change. Manual handling of these stages can make releases inconsistent and increase the effort required to reproduce and update systems.

The internship therefore addressed the broader problem of understanding how AI applications can be prepared for production through deployment and automation practices. This includes API-based model serving, containerization, CI/CD, cloud deployment, Generative AI serving, agentic workflows, security and compliance, orchestration, monitoring, optimization, model drift, retraining, and RAG-based applications.

### **4.4 Project Objectives**

1. Understand the AI lifecycle from development and validation to deployment and operation.
2. Learn how FastAPI can be used for model serving and application integration.
3. Understand Docker-based containerization and its role in consistent deployment environments.
4. Understand MLOps principles and the purpose of CI/CD pipelines for AI workflows.
5. Study cloud AI deployment concepts using Amazon SageMaker, Google Vertex AI, and Azure Machine Learning.

6. Understand the deployment and serving challenges of Generative AI and LLM applications.
7. Explore LangChain, LlamaIndex, and LangGraph for AI agents and workflow orchestration.
8. Understand AI-powered automation workflows for multi-step business processes.
9. Study AI integration using Streamlit and Gradio-style applications.
10. Understand AI security, privacy, compliance, and bias-mitigation considerations.
11. Learn Kubernetes concepts and production observability with Prometheus and Grafana.
12. Study model performance optimization, cloud cost management, drift detection, retraining, advanced RAG, and vector databases.
13. Consolidate learning through full-stack RAG, AI-agent automation, and end-to-end AI solution projects.

#### **4.5 Scope of the Project**

The scope of the internship covered the full AI deployment lifecycle at a learning and project level. It began with production readiness and API serving, then expanded to packaging, CI/CD, cloud deployment, Generative AI serving, agents, application integration, security, Kubernetes, monitoring, optimization, drift management, RAG, and capstone-oriented implementation.

The scope also included learning how the different components interact. For example, an AI model can be wrapped in a FastAPI service, packaged in Docker, validated in a CI/CD pipeline, deployed to a cloud or Kubernetes environment, connected to a frontend, and then monitored for health and performance. RAG and agentic workflows add retrieval, tools, orchestration, and decision logic to this deployment stack.

Because the internship was delivered as a virtual structured learning program, this report distinguishes curriculum exposure and project work from claims about commercial production systems. Only activities and technologies explicitly supported by the official internship documents are presented as part of the internship.

#### **4.6 Technologies and Tools Used**

| <b>Technology / Tool</b> | <b>Use / Learning Purpose</b>                            |
|--------------------------|----------------------------------------------------------|
| Python                   | Primary language for AI and application development.     |
| FastAPI                  | API deployment and AI model serving.                     |
| Docker                   | Containerization, Dockerfiles, networking, and volumes.  |
| MLOps / CI-CD            | Automation of testing, builds, and deployment workflows. |
| Amazon SageMaker         | Managed cloud AI/ML deployment concepts.                 |

Google Vertex AI  
 Azure Machine Learning  
 vLLM  
 Model Quantization  
 LangChain  
 LlamaIndex  
 LangGraph  
 Streamlit  
 Gradio  
 Kubernetes  
 Prometheus  
 Grafana  
 Evidently AI  
 RAG / Vector Databases  
 Git / Version Control

Managed cloud AI/ML deployment concepts.  
 Managed cloud AI/ML deployment concepts.  
 Serving concepts for large language model workloads.  
 Model optimization for resource-efficient serving.  
 Agent and workflow development.  
 Data-connected AI and agentic application patterns.  
 Agent orchestration and structured workflows.  
 Rapid AI application interface integration.  
 Interactive model and AI demonstrations.  
 Container orchestration using Pods, Deployments, and Services.  
 Monitoring and metric collection.  
 Monitoring dashboards and visualization.  
 Model monitoring and drift-detection concepts.  
 Retrieval-augmented generation and vector retrieval.  
 Tracking source changes and supporting repeatable workflows.

**Table 4.2: Technologies and Tools Covered**

## 4.7 System Architecture

The internship can be represented through a layered AI deployment architecture in which model or application code is moved through validation, packaging, delivery, and operational stages. The design separates development, API serving, deployment, and observability so that each stage can be tested and managed independently.

**Figure 4.1: AI Deployment and Automation Workflow**

| Code / Model | API Layer | Tests | Container | CI/CD | Cloud / K8s | Monitoring | User / App |
|--------------|-----------|-------|-----------|-------|-------------|------------|------------|
|--------------|-----------|-------|-----------|-------|-------------|------------|------------|

In the above workflow, the AI model or application is first prepared as source code and configuration. The API layer provides a service interface, tests validate expected behaviour, Docker provides a reproducible package, and CI/CD automates the path toward deployment. The runtime can be a cloud AI platform or Kubernetes environment. Monitoring and logging provide operational visibility, while a frontend or other client consumes the AI service.

**Figure 4.2: Generic AI Service Deployment Architecture**

| AI Model / LLM           | FastAPI / Application Service | Frontend / Client            |
|--------------------------|-------------------------------|------------------------------|
| Model Artifacts          | Docker Container              | Streamlit / Gradio / Web App |
| Configuration            | Dependencies                  | Prometheus / Grafana / Logs  |
| Model Storage / Registry | Cloud Runtime / Kubernetes    |                              |

This logical architecture also supports modern Generative AI and RAG applications. In a RAG system, the application includes a retrieval layer or vector database that supplies external context to the language model. In an agentic application, additional tool and orchestration components can be placed between the user input and the model or application services.

## 4.8 Methodology

The methodology followed the official eight-week internship structure. Each module built on previous learning and expanded the deployment lifecycle from basic production readiness to advanced, end-to-end AI engineering.

| Week   | Module                                            | Coverage                                                                                                                                                                                   |
|--------|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Week 1 | AI Deployment Foundations & FastAPI Model Serving | AI lifecycle, production readiness, MLOps terminology, environment setup, advanced FastAPI deployment, model integration, and asynchronous operations.                                     |
| Week 2 | Containerization & CI/CD for MLOps                | Docker architecture, Dockerfiles, networking, volumes, MLOps principles, and CI/CD pipelines for machine-learning workflows.                                                               |
| Week 3 | Cloud AI Deployment & Generative AI Serving       | Amazon SageMaker, Google Vertex AI, Azure ML, Generative AI applications, LLM challenges, vLLM serving, and model quantization.                                                            |
| Week 4 | AI Agents & Automation Workflows                  | LangChain, LlamaIndex, LangGraph, agent orchestration, and AI-powered hyper-automation workflows.                                                                                          |
| Week 5 | AI Integration, Security & Compliance             | Streamlit and Gradio integration, AI security, threats, privacy, and bias mitigation strategies.                                                                                           |
| Week 6 | Kubernetes & Observability in Production AI       | Kubernetes Pods, Deployments, Services, Prometheus, Grafana, monitoring, logging, and observability.                                                                                       |
| Week 7 | Performance Optimization, Drift & Advanced RAG    | Model tuning, cloud cost optimization, Evidently AI drift detection, retraining, advanced RAG, agents, and vector databases.                                                               |
| Week 8 | Applied AI Projects & Capstone Projects           | E-commerce and healthcare case studies, full-stack RAG application, AI agent for business automation, and end-to-end production AI solution including deployment, monitoring, and scaling. |

**Table 4.1: Week-wise Internship Learning Structure**

#### **4.8.1 Week 1 - AI Deployment Foundations & FastAPI Model Serving**

The first week established the foundation for production-oriented AI systems. The learning introduced the AI lifecycle, production readiness, MLOps terminology, and environment setup. The module then moved to advanced API deployment using FastAPI, including model integration and asynchronous operations. The main takeaway was that a model becomes an application component only after it is exposed through a reliable service interface.

#### **4.8.2 Week 2 - Containerization & CI/CD for MLOps**

The second week introduced Docker architecture, Dockerfiles, networking, and volumes. Containerization provides a consistent way to package an AI service with its dependencies. The same module introduced MLOps and CI/CD principles, showing how automated pipelines can validate code, build artifacts, and support repeatable delivery of AI workloads.

#### **4.8.3 Week 3 - Cloud AI Deployment & Generative AI Serving**

The third week covered managed cloud AI deployment using Amazon SageMaker, Google Vertex AI, and Azure Machine Learning. It also introduced Generative AI deployment, LLM serving challenges, vLLM, and model quantization. The learning highlighted that larger models introduce additional resource, latency, and efficiency considerations.

#### **4.8.4 Week 4 - AI Agents & Automation Workflows**

The fourth week focused on AI agents and workflow automation. LangChain and LlamaIndex were covered for agent and data-connected application development, while LangGraph addressed orchestration. The goal was to understand how AI can perform multi-step actions and participate in business workflows rather than only generate standalone responses.

#### **4.8.5 Week 5 - AI Integration, Security & Compliance**

Week 5 addressed application integration using Streamlit and Gradio and paired this with security and compliance topics. The curriculum included handling threats, protecting privacy, and mitigating bias. The key lesson was that AI deployment requires both technical integration and responsible operational controls.

#### **4.8.6 Week 6 - Kubernetes & Observability in Production AI**

The sixth week introduced Kubernetes orchestration through Pods, Deployments, and Services. Prometheus and Grafana were used in the observability portion of the curriculum for monitoring and logging concepts. This linked deployment with ongoing operational visibility, which is necessary for diagnosing failures and understanding service health.

#### **4.8.7 Week 7 - Performance Optimization, Drift & Advanced RAG**

The seventh week addressed model performance tuning and cloud cost optimization. It also introduced model drift detection through Evidently AI and retraining strategies. Advanced RAG architectures, LlamaIndex agents, and vector database integrations connected retrieval, generation, and agentic workflows to the broader deployment lifecycle.

#### **4.8.8 Week 8 - Applied AI Projects & Capstone Projects**

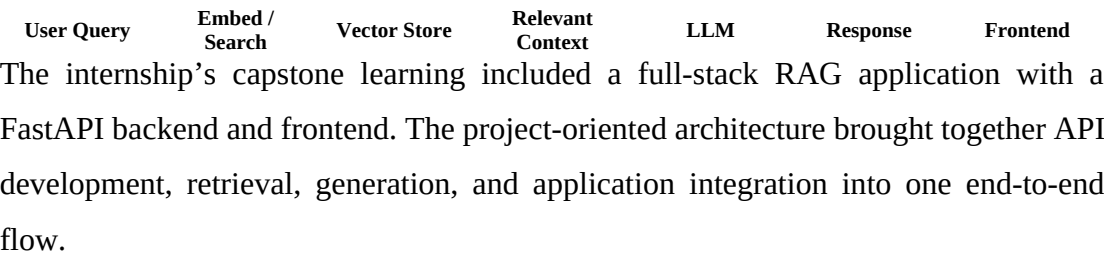
The final week translated the earlier topics into applied projects and case studies. The official curriculum identifies e-commerce and healthcare case studies, a full-stack RAG application with FastAPI backend and frontend, an AI agent for business automation,

and a complete end-to-end AI solution including model development, API deployment, monitoring, and scaling.

4.8.9 Full-Stack RAG Application

Retrieval-Augmented Generation (RAG) connects information retrieval with language generation. A typical RAG workflow accepts a user query, retrieves relevant information from a document store or vector database, and provides the retrieved context to a language model before generating the final response. This helps an application ground generation in an external knowledge source rather than relying only on the model’s internal parameters.

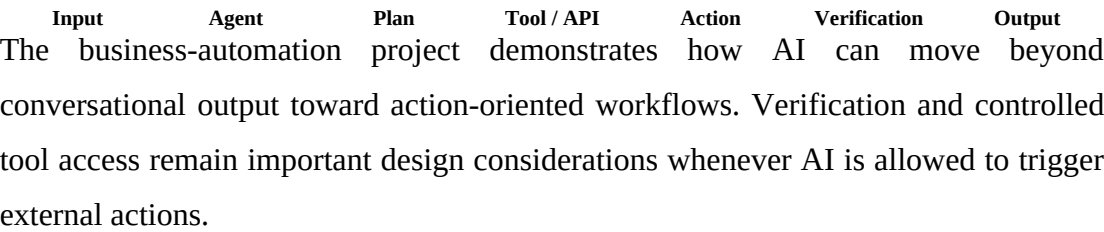
Figure 4.4: RAG Application Flow



4.8.10 AI Agent for Business Automation

An AI agent is a system that combines a language model with tools, memory, retrieval, rules, or external services to perform a multi-step task. Business automation can use an agent to interpret an objective, choose or invoke tools, perform actions, and return a result. The internship curriculum connected this idea with LangChain, LlamaIndex, and LangGraph, emphasizing agent orchestration and AI-powered hyper-automation workflows.

Figure 4.5: AI Agent Automation Workflow



4.8.11 CI/CD Methodology for AI Deployment

A CI/CD pipeline provides a repeatable path from source-code change to deployed service. In an AI setting, the pipeline may include source validation, automated tests, dependency checks, container image building, artifact publishing, deployment, and post-deployment health checks. The internship curriculum introduced these principles in the context of MLOps and Docker-based packaging.

Figure 4.3: CI/CD Pipeline for an AI Application

**Commit      Validate      Test      Build Image      Publish      Deploy      Monitor**

Automation helps reduce variation between releases. When testing and packaging are part of the same workflow, new versions can be evaluated using a consistent process instead of relying entirely on manual deployment commands.

#### 4.8.12 Key Project Components

| Component                  | What Was Covered                              | Learning Purpose                                          |
|----------------------------|-----------------------------------------------|-----------------------------------------------------------|
| FastAPI Model Serving      | API-based model or AI service exposure        | Connect AI logic to application clients.                  |
| Docker Containerization    | Portable packaging of code and dependencies   | Improve consistency across runtime environments.          |
| CI/CD                      | Automated validation, build, and delivery     | Reduce repetitive release work and improve repeatability. |
| Cloud AI Deployment        | Managed cloud deployment concepts             | Understand scalable AI hosting patterns.                  |
| Generative AI Serving      | LLM deployment challenges, vLLM, quantization | Understand compute and serving constraints.               |
| AI Agents                  | Tool-using, orchestrated workflows            | Automate multi-step business processes.                   |
| RAG Application            | Retrieval plus generation                     | Ground responses in external knowledge.                   |
| Kubernetes & Observability | Container orchestration, metrics, logs        | Operate and monitor AI services.                          |
| Drift & Optimization       | Tuning, costs, drift detection, retraining    | Maintain AI quality over time.                            |

**Table 4.3: Key Project Components and Roles**

#### 4.8.13 Learning Outcomes and Skills Developed

| Skill Area                 | Learning Outcome                                                              |
|----------------------------|-------------------------------------------------------------------------------|
| AI Deployment Foundations  | Understanding the AI lifecycle, production readiness, and MLOps terminology.  |
| API Development            | Understanding FastAPI for model serving and application integration.          |
| Containerization           | Understanding Dockerfiles, images, networking, and repeatable environments.   |
| CI/CD                      | Understanding automated validation, building, and deployment.                 |
| Cloud AI                   | Familiarity with SageMaker, Vertex AI, and Azure Machine Learning concepts.   |
| Generative AI              | Understanding LLM serving, vLLM concepts, and model quantization.             |
| Agentic AI                 | Understanding agents, orchestration, tools, and automation workflows.         |
| Responsible AI             | Awareness of security, privacy, compliance, threats, and bias mitigation.     |
| Kubernetes & Observability | Understanding Pods, Deployments, Services, monitoring, and logging.           |
| Optimization & Lifecycle   | Understanding cost management, drift detection, retraining, and advanced RAG. |

**Table 4.4: Learning Outcomes and Skills Developed**

#### 4.8.14 Challenges and Learning Responses

| Challenge                          | Learning Response                                                                                                                                   |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Breadth of the curriculum          | The internship covered multiple deployment and AI technologies. The week-wise sequence helped connect foundational topics to advanced applications. |
| Linking development and operations | The learning emphasized the relationship between models, APIs, containers, CI/CD, cloud runtimes, monitoring, and maintenance.                      |
| Serving larger AI models           | LLM serving can create resource and latency constraints. vLLM and quantization were studied as relevant serving and optimization concepts.          |



|                         |                                                                                                                                                      |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Complex agent workflows | Agent orchestration introduces multiple components. LangGraph and structured workflows were used as conceptual tools for organizing agent behaviour. |
| Production visibility   | Prometheus, Grafana, monitoring, and logging concepts were studied to understand operational observability.                                          |
| Model lifecycle change  | Drift detection and retraining strategies were covered to understand how deployed models may require maintenance as conditions change.               |

**Table 4.5: Challenges and Learning Responses**

#### **4.8.15 Assessment and Internship Structure**

The official offer letter states that the internship followed a week-wise structured learning plan. Weekly assessments were used to track progress and understanding, and certain modules required project documentation and assignments. The final week included a Final Assessment Test, and the final grade was based on overall performance including weekly assessments, project submissions, and the final test.

The EduSkills internship portal also describes a broader process involving application, verification and offer-letter issuance, an 8-week learning module, project submission and credential validation, final assessment, and certification after successful completion.

### **4.9 Expected Outcomes**

- Ability to explain the AI deployment lifecycle and the role of MLOps.
- Ability to describe API-based AI serving using FastAPI.
- Understanding of Docker packaging and consistent runtime environments.
- Understanding of CI/CD concepts for AI and machine-learning workflows.
- Awareness of managed cloud AI platforms and deployment patterns.
- Understanding of Generative AI serving challenges and optimization approaches.
- Knowledge of AI-agent frameworks and automation workflows.
- Awareness of security, privacy, compliance, and bias-mitigation requirements.
- Understanding of Kubernetes concepts and production observability.
- Understanding of model optimization, cost management, drift detection, retraining, and RAG.
- Practical project exposure to a full-stack RAG application and AI-agent business automation.
- Improved ability to discuss an end-to-end production-oriented AI architecture.

#### **4.10 Certificates of Completion and Other Communication Mail Proof**

The official internship completion certificate is attached as Appendix A. It confirms successful completion of the 8-week AI Deployment & Automation program and lists the competencies demonstrated during the internship.

The official internship offer letter is attached as Appendix B. It records the internship reference number, student details, domain, duration, AICTE student ID, structured week-wise curriculum, assessment method, and the requirement for successful completion before the certificate is awarded.

No separate e-mail communication proof was supplied with the documents used for this report. If the university requires e-mail proof, it should be attached separately before final submission.

## **5. BIBLIOGRAPHY / REFERENCES**

1. EduSkills Foundation. "About - EduSkills Foundation." Official website. <https://eduskillsfoundation.org/index.php/about/>
2. EduSkills Foundation. "AICTE-EduSkills Internship." Official internship portal. <https://eduskillsfoundation.org/internshipreg/>
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9. Google Cloud. Vertex AI Documentation.
10. Microsoft Learn. Azure Machine Learning Documentation.
11. Kubernetes Documentation. <https://kubernetes.io/docs/>
12. Prometheus Documentation. <https://prometheus.io/docs/>
13. Grafana Documentation. <https://grafana.com/docs/>
14. Evidently AI. Documentation for machine-learning monitoring and drift detection.
15. LangChain Documentation. <https://python.langchain.com/>
16. LlamaIndex Documentation. <https://docs.llamaindex.ai/>

## APPENDIX A - OFFICIAL INTERNSHIP CERTIFICATE

Official certificate supplied for the AICTE - EduSkills Virtual Internship in AI Deployment & Automation.

**CERTIFICATE  
OF INTERNSHIP**





**PRESENTED TO**

***Toshi Pandey***

*has successfully achieved the Internship Credential by completing a 8-week program in the domain of*

**AI Deployment & Automation**

*During the internship, the learner demonstrated proficiency in:*

- Foundations of AI Deployment & Production Readiness & Advanced API Deployment with FastAPI
- Containerization with Docker & Introduction to MLOps & CI/CD Pipelines
- Cloud Deployment Basics (AWS SageMaker, Google Vertex AI, Azure ML) & Deploying Generative AI Applications
- Building Advanced AI Agents with LangChain & LlamaIndex & AI-Powered Automation Workflows
- Integrating AI into Web/Mobile Apps & Security and Compliance in AI Deployment
- Container Orchestration with Kubernetes & Monitoring, Logging & Observability
- Performance Optimization & Cost Management & Model Drift Detection & Retraining & Advanced RAG & Agentic Workflows
- Case Studies & Best Practices in AI Deployment & Project: Deploy a Full-Stack RAG Application & Project: AI Agent for Business Automation & Capstone Project: End-to-End AI Solution for Business Problem

This certification recognizes the successful completion of the internship requirements.



Issued on: Sep 10, 2026  
Certificate ID: 2026-D1DDA7AB3A



**Mitu Swain**  
General Manager  
(Learning & Development)  
EduSkills



## APPENDIX B - INTERNSHIP OFFER LETTER

Official internship offer letter supplied by EduSkills Foundation.



Ref No: C17Y2026M091505270

### Internship Offer Letter

**Student Name** : Toshi Pandey  
**Stream** : B.Tech  
**Branch** : B.Tech(Computer Science and Engineering)  
**Institute Name** : IILM University, Greater Noida  
**Internship Domain** : AI Deployment & Automation  
**Internship Duration** : June 2026 - August 2026  
**AICTE Student ID** : STU6999545eb0d751771656286

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

#### **Internship Structure & Evaluation:**

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

**The week wise Structure of the Internship is as Follows:**

| Week                                     | Module                                            | Description                                                                                                                                                                                                                                                                    |
|------------------------------------------|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Week 1                                   | AI Deployment Foundations & FastAPI Model Serving | Introduction to AI lifecycle, production readiness, MLOps terminology, environment setup, and advanced API deployment using FastAPI including model integration and asynchronous operations.                                                                                   |
| Week 2                                   | Containerization & CI/CD for MLOps                | Docker architecture, Dockerfiles, networking, volumes, introduction to MLOps principles, and implementing CI/CD pipelines for machine learning workflows.                                                                                                                      |
| Week 3                                   | Cloud AI Deployment & Generative AI Serving       | Deployment on cloud platforms like Amazon SageMaker, Google Vertex AI, and Azure Machine Learning, along with deploying Generative AI applications, handling LLM challenges, vLLM serving, and model quantization.                                                             |
| Week 4                                   | AI Agents & Automation Workflows                  | Building advanced AI agents using LangChain and LlamaIndex, agent orchestration with LangGraph, and implementing AI-powered hyper-automation workflows.                                                                                                                        |
| Week 5                                   | AI Integration, Security & Compliance             | Integrating AI into web and mobile apps using Streamlit and Gradio demos, securing AI systems, handling threats, ensuring privacy, and bias mitigation strategies.                                                                                                             |
| Week 6                                   | Kubernetes & Observability in Production AI       | Container orchestration using Kubernetes, working with Pods, Deployments, Services, and implementing monitoring & logging with Prometheus and Grafana.                                                                                                                         |
| Week 7                                   | Performance Optimization, Drift & Advanced RAG    | Model performance tuning, cloud cost optimization, model drift detection using Evidently AI, retraining strategies, advanced RAG architectures, LlamaIndex agents, and vector database integrations.                                                                           |
| Week 8                                   | Applied AI Projects & Capstone Projects           | Real-world case studies (e-commerce & healthcare), full-stack RAG application development (FastAPI backend + frontend), AI agent for business automation, and complete end-to-end production AI solution including model development, API deployment, monitoring, and scaling. |
| <b>Internship Grade Point Assessment</b> |                                                   |                                                                                                                                                                                                                                                                                |

**Note: There is no stipend associated with this internship.**

We congratulate you on your selection and wish you a successful learning journey.



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